Department of Physics

PCS 125 – WAVES AND FIELDS

MOCK EXAM

1. A 100 g mass hanging from a 1.0 m long string is pulled 8.0° to one side and released. How long does it take for the pendulum to reach 4.0° on the opposite side?

   (a) 1.8 s (b) 0.67 s (c) 1.1 s (d) 0.33 s (e) 0.56 s

2. A 2.0 g spider is dangling at the end of a silk thread. You can make the spider bounce up and down on the thread by tapping lightly on his feet with a pencil. You soon discover that you can give the spider the largest amplitude on his little bungee cord if you tap exactly once every second. What is the spring constant of the silk thread?

   (a) 0.079 N/m (b) 0.79 N/m (c) 1.07 N/m (d) 0.039 N/m (e) 1.56 N/m

3. A traveling wave is described by the function in the xy plane:

$$y(x,t) = 0.1 \cos [- 4.0\, x + 3.0\, t + 3.0]$$

where x and y are in meters and t is in seconds. What is the maximum transverse velocity of a point in the medium?

(a) 0.3 m/s **j** (b) 0.75 m/s **i** (c) 0.3 m/s **i** (d) 0.75 m/s **j**

(e) other (give value)

4. A sinusoidal sound wave described by the displacement wave function $s(x, t) = 2.00 \cos(15.7x - 858t)$ where s is in micrometers, x is in meters, and t is in seconds, moves through a medium.
Determine the instantaneous speed of an element of the medium at $x = 0.050$ m and $t = 3.00$ ms.

(a) 54.7 m/s (b) –54.7 μm/s (c) 1.67 mm/s (d) –1.67 m/s

(e) other (give value – 1.67 mm/s)

5. What is the ratio of the tension in a guitar string tuned to E-flat (f = 310 Hz) to that in a string tuned to E (f = 330 Hz)? The guitar strings have the same length and thickness.

(a) 1.1 (b) 0.94 (c) 0.88 (d) 0.75 (e) 0.11

6. An experimenter wishes to generate in air at 39^0C a sound wave that has a displacement amplitude of 5.5×10^{-6} m. The pressure amplitude can not exceed 0.840 N/m^2. What is the minimum wavelength the sound wave can have (Given ρ_{air} =1.20kg/m^3) ?

(a) 5.81 m (b) 6.18 m (c) 0.17m (d) 7.06 m (e) other (give value)

7. The power output of a public-address speaker is 8.00 W. Suppose it broadcasts equally in all directions. If the air medium absorbs no energy, at what minimum distance from the speaker would the sound be barely audible for a normal person?

(a) 691 km (b) 637 km (c) 798 km (d) 125 km (e) 157 km

8. Two transverse sinusoidal waves are described by the wave functions
$y_1 = 15.0 \sin [\pi(x + 0.200t)/2]$ and
$y_2 = 15.0 \sin [\pi(x - 0.200t)/2]$, where x is in meters, y_1, and y_2 are in centimeters and t is in seconds. Determine the maximum transverse position of an element of the medium at x = 1.80 m due to the superposition of the two waves.

(a) 9.27 cm (b) 4.64 cm (c) 8.82 cm (d) 17.6 cm (e) 1.48 cm

9.
The variation in pressure of helium gas, measured from its equilibrium value, is given by $\Delta P = 2.9\times10^{-5}\cos(4.2x - 3000t)$ where x and t have units of m and s respectively and ΔP is measured in Pa. Determine the wavelength of the wave.

(a) 0.5 m (b) 0.67 m (c) 1.49 (d) 477 m (e) 239 m

10. To decrease the intensity of the sound you are hearing from your speaker system by a factor of 36, you can

(a) reduce the amplitude by a factor of 12 and increase your distance from the speaker by a factor of 3.
(b) reduce the amplitude by a factor of 4 and increase your distance from the speaker by a factor of 3.
(c) reduce the amplitude by a factor of 2 and increase your distance from the speaker by a factor of 3.
(d) reduce the amplitude by a factor of 3 and increase your distance from the speaker by a factor of 4.
(e) Other (give alternative)

11. Determine the escape speed from a planet of mass M and radius R with $M = 3.2 \times 10^{23}$ kg and

$R = 2.4 \times 10^{6}$ m.

(a) 5.5 km/s (b) 4.2 km/s (c) 5.2 km/s (d) 4.8 km/s (e) 3.7 km/s

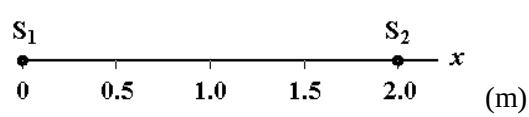
12. The planet Venus requires 225 days to orbit the sun ($M_{sun} = 1.99 \times 10^{30}$ kg), in an almost circular trajectory. Calculate the radius of the orbit and the orbital speed of Venus as it circles the sun.

(a) 2.10×10^{9} m, 44.2 km/s (b) 1.08×10^{9} m, 12.2 km/s

(c) 1.08×10^{11} m, 34.9 km/s (d) 2.10×10^{9} m, 34.9 km/s

(e) None of the above

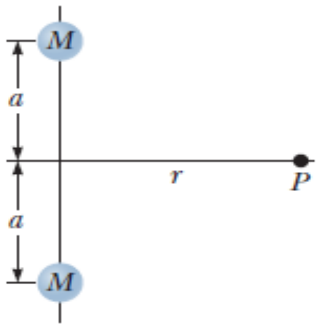
13. Two point sources emit sound waves of 1.0-m wavelength. The sources S_1 and S_2 which are at a distance of 2.0 m from each other (cf. the figure given) emit waves which are in phase at the instant of emission. Where, along the line between the two sources, are the waves out of phase by π radians with each other?



(a) x = 0, 1.0 m, 2.0 m
(b) x = 0.50 m, 1.5 m
(c) x = 0.50 m, 1.0 m, 1.5 m
(d) x = 0.75 m, 1.25 m

(e) x = 0.25 m, 0.75 m, 1.25 m, 1.75 m

14. Two point objects of equal mass M each are separated by a distance $2a$, as shown in the figure below. What is the magnitude of the gravitational force on a 1.0 kg mass if placed at P?



a. $\dfrac{2MGr}{\sqrt{(a^2+r^2)^3}}$

b. $\dfrac{2MGa}{\sqrt{(a^2+r^2)^3}}$

c. $\dfrac{MGr}{\sqrt{(a^2+r^2)^3}}$

d. $\dfrac{MGa}{\sqrt{(a^2+r^2)^3}}$

e. $\dfrac{2MGr}{(a^2+r^2)}$

15. A device meter to measure blood-flow speed uses Doppler effect with ultrasound waves. Suppose the device emits waves of frequency 3.5 MHz at a large leg artery. What is the expected beat frequency between the emitted wave and the wave received after being reflected from the blood if the blood is flowing at 2.0 cm/s directly away from the sound device? Assume the speed of sound in human tissue to be 1540 m/s.

(a) 91 Hz

(b) 25 Hz

(c) 45 Hz

(d) 250 Hz

(e) other (give value)

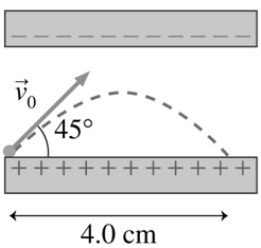
16. A satellite orbits the earth with constant speed at height above the surface equal to half of the earth's radius. The magnitude of the satellite's acceleration is expressed in terms of the acceleration of gravity on earth, g, by

(a) $g/4$
(b) $g/2$
(c) 4g/9
(d) 9g/4
(e) $3g/2$

17. An electron is placed at each corner of an equilateral triangle of sides 20 cm long. What is the electric field at the midpoint of one of the sides?

(a) 4.8×10^{-8} N/C (b) 1.05×10^{-8} N/C (c) 6.0×10^{-9} N/C (d) 2.0×10^{-9} N/C (e) 3.5×10^{-8} N/C

18. An electron is launched at a 45° angle with a speed of 7.0×10^{6} m/s from the positive plate of the parallel plates shown in the figure below. The electron lands 4.0 cm away. Determine the electric field in the space between the plates.



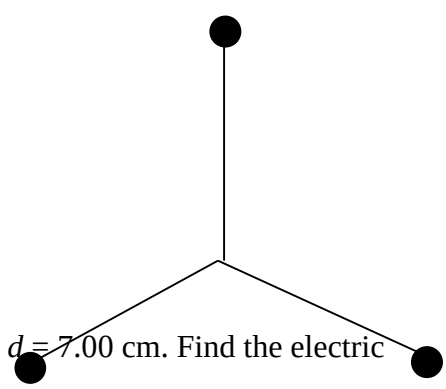
(a) 0.53×10^{4} N/C (b) 3.56×10^{3} N/C (c) 1.23×10^{4} N/C
(d) 6.98×10^{3} N/C (e) None of these

19. Two identical conducting small spheres are placed with their centers 0.30 m apart. One is given a charge of 12.0 nC and the other a charge of −18.0 nC. If the spheres are connected by a conducting wire, determine the electric force each sphere exerts on the other after they have reached electrostatic equilibrium?

(a) 7.99×10^{-7} N (b) 8.99×10^{-7} N (c)8.99×10^{-9} N (d) 6.77×10^{-9} N (e) 8.25×10^{-7} N

20. Three 2.5 μC charges are placed on tiny conducting spheres at the ends of 1.0 m-long strings that are connected at 120° angles, as shown below. The three charges are on the *xy*-plane. The magnitude of the tension in any one of the strings is

(a) 1.9×10^{-2} N
(b) 7.5×10^{-2} N
(c) 3.7×10^{-2} N
(d) 6.5×10^{-2} N
(e) Other (give value) 9.74×10^{-2} N



21. Two charges (q_1 = –26.0 nC and q_2 = 11.5 nC) are separated by d = 7.00 cm. Find the electric potential at a point located 7.00 cm from q_1 and 7.00 cm from q_2.

(a) –1.86 kV (b) 1.86 kV (c) –4.82 kV (d) 4.82 kV (e) 0 V

22. Read carefully the following 4 statements. Which are false?

A) The work done by an external agent to separate two masses m_1, m_2 which are at a distance r to an infinite distance from each other is: $G\frac{m_1 m_2}{r}$.

B) The work done by an external agent to separate two masses m_1, m_2 which are at a distance r to an infinite distance from each other is: $-G\frac{m_1 m_2}{r}$.

C) The work done by the electric field to separate two charges of equal sign q_1, q_2 which are at a distance r to an infinite distance from each other is: $-k_e\frac{q_1 q_2}{r}$.

D) The work done by the electric field to separate two charges of equal sign q_1, q_2 which are at a distance r to an infinite distance from each other is: $k_e\frac{q_1 q_2}{r}$.

(a) A only (b) B only (c) A and C (d) B and D (e) none of the above B and C are false

23. A particle with positive charge q = 3.20 ×10^{-19} C moves with a velocity $\vec{v} = (2\hat{i} + 3\hat{j} - \hat{k})$ m/s through a region where both a uniform magnetic field and a uniform electric field exist. Calculate the total force on the moving particle, when $\vec{B} = (2\hat{i} + 4\hat{j} + \hat{k})$ and $\vec{E} = (4\hat{i} - \hat{j} - 2\hat{k})$ V/m .

(a) $(3.52\hat{i} + 1.60\hat{j}) \times 10^{-18}$ N

(b) $(3.52\hat{i} - 1.60\hat{j}) \times 10^{-18}$ N

(c) $(1.60\hat{i} + 3.52\hat{j}) \times 10^{-18}$ N

(d) $(1.60\hat{i} - 3.52\hat{j}) \times 10^{-18}$ N

(e) $(9.6\hat{i} - 3.2\hat{j} + 38.4\hat{k}) \times 10^{-19}$ N

24. A particle of mass m= 5.0 g and charge q= 40 mC moves in a region of space where the electric field is uniform and is given by E_x = 5.5 N/C, E_y = E_z = 0. If the position and velocity of

the particle at $t = 0$ are given by $x = y = z = 0$ and $v_x = 50$ m/s, $v_y = v_z = 0$, what is the distance from the origin to the particle at $t = 2.0$ s?

(a) 60 m

(b) 28 m

(c) 44 m

(d) 188 m

(e) Other (give value)

25. A Hall-effect probe uses a semiconductor with a charge carrier density of 7.5×10^{20} m^{-3}. The probe has a cross section area of 1.23×10^{-5} m^2 and its dimension in the direction of the magnetic field being measured is 0.35 mm. The probe carries a 2.5 mA current perpendicular to the magnetic field. If its Hall potential is 4.5 mV, what is the magnetic field strength? (Assume the charge of the carriers is the electron charge)

(a) 53 mT (b) 5.3 mT (c) 33.3 mT

(d) 76 mT (e) 18.8 mT

26. A wire carries a 3.0A current along the *x*-axis through a magnetic field $\vec{B} = (3.0\hat{i} + 9.0\hat{j})$ T. What is the length of the wire if the wire experiences a force of $23.0\hat{k}$ N as a result of being in the magnetic field?

(a) 0.81 m (b) 2.6 m

(c) 0.64 m (d) 0.85 m

(e) 1.28 m

27. The boundary shown in the figure is that of a uniform magnetic field directed in the positive *z* direction (out of the paper). An electron enters the magnetic field at the origin with a velocity in the positive *x* direction and exits 0.263 μs later at point A. What is the magnitude of the magnetic field?

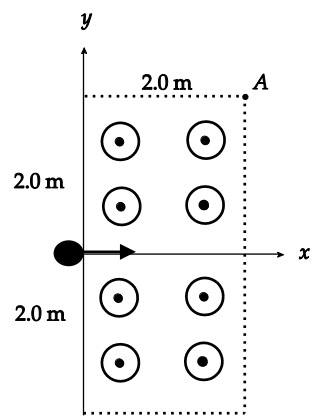
a. 18 μT

b. 34 μT

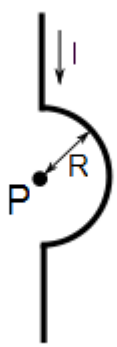
c. 28 μT

d. 14 μT

e. 227 μT



28. Find the magnetic field at point P located at the center of a semicircular current loop of radius R = 20 cm when the current is I = 15A.



(a) 2.4×10^{-5} T, into the page (b) 2.4×10^{-5} T, out of the page

(c) 4.7×10^{-5} T, into the page (d) 4.7×10^{-5} T, out of the page

(e) None of the above

29. A long straight wire carries a current I along the x-axis. What is the direction of the force on a second long wire in the x-y plane, with current in the same direction, when it is placed first at $y = +10$ cm and then at $y = -10$ cm?

(a) Up (+y), then Down (-y)

(b) Up, then Up

(c) Down, then Up

(d) Down, then Down

(e) No force at either position (currents are parallel).

30. A metal strip has dimensions 7 cm × 4 cm × 2 cm in the $x \times y \times z$ directions respectively and is placed in a magnetic field. Which of the following statements is true?

Statement A: If the current in the metal strip is in the $+y$ direction, the magnetic field is in the $+x$ direction and the Hall electric field is in the +z direction, then the carriers are holes (positive particles).

Statement B: If the current in the metal strip is in the $+x$ direction, the magnetic field is in the $+y$ direction and the Hall electric field is in the -z direction, then the carriers are electrons.

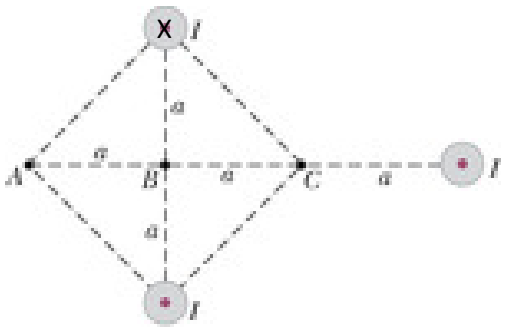
Statement C: If the current in the metal strip is in the $+x$ direction, the magnetic field is in the $+z$ direction and the Hall electric field is in the $-y$ direction, then the carriers are electrons.

(a) only statement A is true
(b) only statement B is true
(c) only statement C is true
(d) statements A and C are true but B is false
(e) statements B and C are true but A is false

31. When two negative charges move towards each other, the electric potential energy of the two charge system

(a) increases
(b) remains the same
(c) decreases
(d) first decreases then increases
(e) first increases then decreases

32. Three long parallel conductors carry currents of I=3.00 A. The figure in this problem is a top view of the conductors with two of the currents coming out of the paper and one into. Taking the dimension a to be a= 2.00 cm, determine the magnitude and direction of the magnetic field at point A.



(a) 40 μT (-**j**)
(b) 40 μT (-**i**)
(c) 30 μT **j**
(d) 0 μT
(e) Other (give value)

33. If 5×10^{21} electrons pass through a 15-Ω resistor in 10 min, what is the potential difference across the resistor?

(a) 10 V
(b) 32 V
(c) 38 V
(d) 20 V
(e) 28 V

34. Two rings of diameter $2d$ are placed a distance d apart, one right above the other as shown. They both carry current I in same direction. What is the magnetic field on the middle point of the line connecting the centers of the rings?

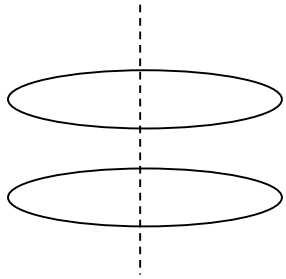
(a) 0

(b) $2\mu_0 I/d$

(c) $\mu_0 I/d$

(d) $0.7\mu_0 I/d$

(e) $1.4\mu_0 I/$



35. A 4.0-W resistor has a current of 3.5 A in it for 6.0 min. How many electrons pass through the resistor during this time interval?

(a) 7.5×10^{21}

(b) 5.6×10^{21}

(c) 6.6×10^{21}

(d) 7.9×10^{21}

(e) Other (give value)

36. A loose spiral spring is hung from the ceiling, and a large current is sent through it. The coils move

(a) closer together

(b) farther apart

(c) they do not move at all

(d) closer together or farther apart, depending on the direction of the current.

(e) closer together or farther apart, depending on the mass of the spring and the spring constant.

37. Two long, parallel wires are attracted to each other by a force per unit length of 320 μN/m when they are separated by a vertical distance of 0.500 m. The current in the upper wire is 20.0 A to the right. Determine the location of the line in the plane of the two wires along which the total magnetic field is zero.

(a) 0.17 m below the upper wire.

(b) 0.17 m above the upper wire.

(c) 0.33 m above the upper wire.

(d) 0.33 m below the upper wire.

(e) Other (give value)

38. A copper bar of square cross section is oriented vertically and has negative charge carriers. The four sides of the bar face north, south, east, and west. There is a uniform magnetic field directed from east to west, and the bar carries current downward. Which side of the wire is at the highest electric potential?

(a) north side
(b) south side
(c) east side
(d) west side
(e) the electric potential is the same on all sides